Answer Key — Competitive Exam Practice 6: Process State Diagram, Schedulers, and Context Switch

GATE (Computer Science) Pattern

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Concepts That Click

Q1 (C). A process whose event occurs goes `waiting` → `ready` and must then be dispatched by the short-term scheduler; there is no direct `waiting` → `running` arrow. (A) is preemption, (B) is event completion, (D) is an I/O request — all legal.

Q2 (B). The long-term (job) scheduler decides which processes are admitted into memory, and so controls n, the number of processes in memory. The short-term scheduler and dispatcher only choose among processes already in memory.

Q3 (C). Swapping out and later swapping in a partially executed process is the medium-term scheduler's job; it reduces the degree of multiprogramming temporarily to improve the mix or free memory.

Q4 (C). An empty pipe forces the caller to block (`running` → `waiting`), so the kernel must switch to another process. `getpid` (A) is a mode switch only. In (B) the woken process merely moves to `ready`; the interrupted process may resume. (D) never enters the kernel.

Q5 (B). A switch to a different process can only be performed by the kernel, so it always involves entering kernel mode (and returning to user mode in the new process): S1 is true. A system call such as `getpid`, or an interrupt that returns to the interrupted process, is a mode switch with no process switch: S2 is false.

Q6 28. Running = 8 (one per core). Waiting = $50 - 8 - 11 - 3 - 0 = 28$.

Q7 10. Overhead = $s/(q+s) = 5/(45+5) = 5/50 = 0.10 = 10\%$.

Q8 24. Require $1/(q+1) \le 0.04 \iff q+1 \ge 25 \iff q \ge 24\,\text{ms}$.

Q9 3. Legal: `new` → `ready`, `running` → `ready`, `waiting` → `ready`. Illegal: `ready` → `waiting` (only a running process can issue a request to wait on), `waiting` → `running` (must pass through `ready`), `new` → `running` (must be admitted to `ready` first).

Q10 3. Timeline: `P1` runs 0–4 (2 ms left); switch to `P2`, runs 4–8 (1 ms left); switch to `P1`, runs 8–10 and terminates; switch to `P2`, runs 10–11 and terminates. Switches at $t = 4, 8, 10$: 3. (The CPU goes idle at $t = 11$; that is not a switch to a different process.)